PRACTICAL NO. 9 – STUDENT MANAGEMENT SYSTEM (LOGIN MODULE)

Aim: To analyze and apply Black-Box Testing, White-Box Testing, Integration Testing, and System Testing on a sample Student Management System (Login Module).

Test Case ID	Testing Type	Scenario	Test Case Description	Preconditions	Test Steps	Test Input	Expected Result	Actual Result	Status	Comments
BB-01	Black-Box Testing	Valid login	Verify login with correct student credentials	Test student account is available; login page is open	1. Open login page 2. Enter valid username and password 3. Click Login	student01 / Valid@123	Student is authenticated and redirected to Student Dashboard	Dashboard is displayed for the sample valid account	Pass (sample)	Positive login path; screenshot 04
BB-02	Black-Box Testing	Invalid password	Verify incorrect password is rejected	Test student account exists	1. Open login page 2. Enter registered username 3. Enter incorrect password 4. Click Login	student01 / Wrong@123	Error is displayed; user remains on login page	Error message 'Invalid username or password' is displayed; dashboard not opened	Pass (sample)	Negative login path; screenshot 02
BB-03	Black-Box Testing	Empty fields	Check validation when username and password are blank	Login page is open	1. Leave username and password blank 2. Click Login	Username: blank; Password: blank	Required-field validation appears or login is prevented	Required-field message is displayed; login is not completed	Pass (sample)	Input validation; screenshot 03
WB-01	White-Box Testing	Valid/invalid credential branches	Review the decision branch for credential matching	Login source code or pseudocode is available for review	1. Inspect credential comparison condition 2. Trace matching branch 3. Trace non-matching branch	Matching and non-matching credential pairs	Matching branch allows access; non-matching branch rejects access	Expected branch behaviour documented; source-code execution not independently verified	Pass (logic review)	White-box result is based on logic review template
WB-02	White-Box Testing	Empty-input condition	Check branch for missing username/password	Login validation logic is available	1. Inspect empty-field condition 2. Test blank username 3. Test blank password	Username blank / Password blank	Empty-input branch returns validation and does not authenticate	Empty-input validation branch is covered in the sample test design	Pass (logic review)	Branch review; see screenshot 03
WB-03	White-Box Testing	Authentication failure handling	Review failure path does not create an authenticated session	Authentication logic is available for review	1. Trace failed authentication branch 2. Confirm no session is created 3. Confirm return to login page	Incorrect credentials	No authenticated session is created; safe error is returned	Failure path is specified; requires source/runtime verification in actual system	Pass (logic review)	Confirm in actual source code before final submission if required
IT-01	Integration Testing	Login and student database	Verify authentication checks student record	Login module is connected to test database	1. Submit registered test student's credentials 2. Check authentication response 3. Verify student record mapping	Registered test student credentials	Correct student record is validated and login succeeds	Sample expected integration flow reaches the student dashboard	Pass (sample flow)	Requires a real database environment for live verification
IT-02	Integration Testing	Login and session/dashboard	Verify successful login creates session and opens dashboard	Authentication, session, and dashboard modules are integrated	1. Log in with valid credentials 2. Check dashboard access 3. Verify authenticated student context	Valid student credentials	Session is created and dashboard displays the correct student context	Sample dashboard shown after valid-login scenario	Pass (sample flow)	Screenshot 04 is an illustrative UI mockup
IT-03	Integration Testing	Login failure response	Verify UI handles authentication rejection	Login UI and authentication service are connected	1. Submit invalid credentials 2. Observe UI response 3. Confirm no dashboard access	Invalid credentials	UI displays suitable error and remains unauthenticated	Sample error message is displayed; dashboard is not shown	Pass (sample flow)	Screenshot 02
ST-01	System Testing	End-to-end successful login	Test the complete successful login workflow	Test system and student test account are available	1. Open login page 2. Enter valid credentials 3. Click Login 4. Verify dashboard	Valid student credentials	User reaches the correct dashboard	Sample workflow reaches Student Dashboard	Pass (sample flow)	Screenshot 01 and 04
ST-02	System Testing	End-to-end failed login	Ensure invalid credentials cannot access protected pages	Login page is available	1. Enter invalid credentials 2. Submit 3. Attempt to open dashboard directly	Invalid credentials	Login is rejected and protected dashboard remains inaccessible	Sample login displays error and does not navigate to dashboard	Pass (sample flow)	Screenshot 02; live access control needs system verification
ST-03	System Testing	Logout/session clearing	Verify logout ends authenticated session	Student is logged in	1. Click Logout 2. Attempt to revisit dashboard using Back button or URL	Active student session	User returns to login page; protected content is inaccessible without login	Logout scenario is documented; session invalidation must be checked on the actual system	Pass (test-plan result)	Confirm by running against deployed system

Screenshot ID	Screenshot / Evidence	What it demonstrates	File included in Word report	Evidence note
SS-01	Student Login Page	Login form and fields	Yes	Illustrative mockup, not live capture
SS-02	Invalid Login	Error for invalid credentials	Yes	Illustrative mockup, not live capture
SS-03	Empty Input Validation	Required-field message	Yes	Illustrative mockup, not live capture
SS-04	Student Dashboard	Example destination after successful login	Yes	Illustrative mockup, not live capture

PRACTICAL NO. 9 – NOTES	
Module	Student Management System – Login Module
Status meaning	Pass (sample) means the illustrative sample flow matches the expected result; it does not prove execution on your institute's live application.
White-Box requirement	Inspect source code or pseudocode to confirm the actual decision branches before submitting as live code-based testing.
Integration requirement	Database, authentication service, session, and dashboard tests need a real connected test environment for live evidence.
Screenshots	Screenshots are illustrative mockups because no live application URL or actual screenshots were supplied. Replace with real screenshots if required by faculty.
Credentials	Use dummy/test credentials only. Never include your personal password in the report.
Submission	Review the tester name, date, URL, and any faculty-specific format requirements before uploading.